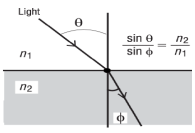
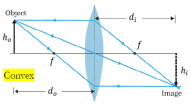
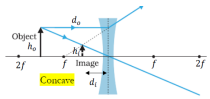
Property of EM radiation	with WAVE	PARTICLE
Reflection	Yes	Yes
Refraction	Yes	Yes
Interference	Yes	No
Diffraction	Yes	No
Polarization	Yes	No
Photoelectric effect	No	Yes



Ray and wave optics n is real
EM wave optics n is comple



Real images
Detectable on a screen
Formed by real rays
Opposite side of object



Virtual images
Cannot be detected on a screen
Formed by extensions.
Same side as object

The properties of a glass flat prism can be determined from geometry.
The prism angle is A.

$\frac{\sin\alpha_i}{\sin\alpha_o} = n \qquad \frac{\sin\beta_i}{\sin\beta_o} = \frac{1}{n}$

$(180-A)+\alpha_o+\beta_i = 180$

$\beta_i = A-\alpha_o = \beta_o - b$

$\alpha_i = a+\alpha_o$

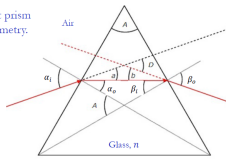
$(180-D)+a+b = 180$

$D = a+b = \alpha_i-\alpha_o+\beta_i-\beta_o$

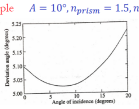
$D = (\alpha_i+\beta_o)-A$

An important parameter is the minimum deviation angle D_{min}.

$\frac{n_{prism}}{n_o} = \frac{\sin[(A+D_{min})/2]}{\sin(A/2)}$



Example $A = 10°, n_{prism} = 1.5, n_o = 1$



Least time to reach target.

$t_1 = \frac{AP}{v_1} = \frac{\sqrt{x^2+a^2}}{v_1}$

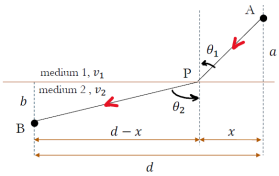
$t_2 = \frac{PB}{v_2} = \frac{\sqrt{(d-x)^2+b^2}}{v_2}$

$t = t_1 + t_2$

$t(x) = \frac{\sqrt{x^2+a^2}}{v_1} + \frac{\sqrt{(d-x)^2+b^2}}{v_2}$

$\frac{dt}{dx} = \frac{\sin\theta_1}{v_1} - \frac{\sin\theta_2}{v_2}$

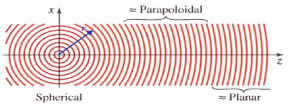
$\frac{dt}{dx} = 0 \Rightarrow \sin\theta_2 = \sin\theta_1 \cdot \frac{v_2}{v_1} = \sin\theta_1 \cdot \frac{n_1}{n_2}$



Snel's Law of Refraction $n_1 \sin\theta_1 = n_2 \sin\theta_2$

Snel's Law of Reflection $\sin\theta_i = \sin\theta_r$

Mirror type	Lens type	f	d_o	d_i	$M = h_i/h_o$	Image
Concave	Convex	+	$d_o > 2f$	$2f > d_i > f$	Reduced inverted	Real
			$2f > d_o > f$	$d_i > 2f$	Enlarged inverted	Real
			$f > d_o > 0$	$\lvert d_i\rvert > d_o$	Enlarged	Virtual
Convex	Concave	-	$d_o > 0$	$f > \lvert d_i\rvert > 0$	Reduced	Virtual



$U(r) = a(r)\cdot e^{j\phi(r)}$

Spherical $U(r) = \frac{A}{r}e^{-jkr}$

Plane wav $U(r) = A\,e^{-jkz}$

Parapoloic waves $U(r) = \frac{A}{z}e^{-jkz}\,e^{-jk\left(\frac{x^2+y^2}{2z}\right)}$

The Lens Equation

A relation between the object and image locations with respect to the center line of the lens can be found geometrically. It works for both lens types.

$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \qquad f = R/2$

Example. A 3-cm object is placed 32 cm away from a convex lens with focus length of 8 cm. Describe the image.

$\frac{1}{8} = \frac{1}{32} + \frac{1}{d_i} \qquad d_i = 11\ cm,$

$h_i = -1\ cm, \quad M = \frac{1}{3}$

Lens type	f	d_o	d_i	M	Image
Convex	+	$d_o > 2f$	$2f > d_i > f$	Reduced inverted	Real
		$2f > d_o > f$	$d_i > 2f$	Enlarged inverted	Real
		$f > d_o > 0$	$\lvert d_i\rvert > d_o$	Enlarged	Virtual

Mirrors

Mirrors are single-sided lenses (other surface is metal-coated). The same lens equation is applicable for mirrors.

Example. A store watch mirror with focus length of 3 m. The store is 7 m deep.

$\frac{-1}{3} = \frac{1}{7} + \frac{1}{d_i} \qquad d_i = -2.1\ m, \quad M = 0.3$

Convex		-	$d_o > 0$	$f > \lvert d_i\rvert > 0$	Reduced	Virtual

The scalar wave equation $\nabla^2 U - \frac{1}{v^2}\frac{d^2U}{dt^2} = 0$ $U(r,t)$ is wave function

Assume that the wave function is time-harmonic $U(r,t) = U(r)\,e^{j\omega t}$

The scalar wave equation becomes $\nabla^2 U(r) + \left(\frac{\omega}{v}\right)^2 U(r) = 0$

$(\nabla^2 + k^2)\,U(r) = 0$

Put $k = \frac{\omega}{v}$ Wave number (rad/m)

$k = \frac{\omega}{v} = \frac{2\pi f}{v} = \frac{2\pi}{\lambda}$

$v = \frac{v_o}{n}$

$k = nk_o$

$\lambda = \frac{\lambda_o}{n}$

A general solution is of the form $U(r) = a(r)\cdot e^{j\phi(r)}$

A general time-harmonic solution is of the form $U(r,t) = a(r)\cdot e^{j\phi(r)}\cdot e^{j\omega t}$

An instantaneous solution is of the form $u(r,t) = a(r)\cdot\cos[\omega t + \phi(r)]$ $\phi(r) = kr$

$u(r,t) = \frac{1}{2}[U(r,t) + U^*(r,t)]$

Spherical waves

$U(r) = \frac{\sqrt{I}}{r}e^{-jkr}$

Plane waves

$U(r) = \sqrt{I}\,e^{-jkz}$

1) Optical intensity of a wave is given by

$I(r) = |U(r)|^2$

Such that

$U(r) = \sqrt{I(r)}\cdot e^{j\phi(r)}$

Instantaneous intensity

$I(r,t) = < U(r,t)\cdot U^*(r,t) >$

2) Wave power is given by

$P = \int_A I(r)\,dA$

Where does the phase come from?

1) Sources are out-of-phase.

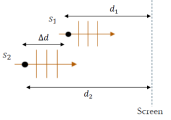
2) Distance.

$U_1 = \sqrt{I_o}\cdot e^{-jkd_1} \qquad U_2 = \sqrt{I_o}\cdot e^{-jkd_2}$

$I = 2I_o[1+\cos(k\Delta d)]$

$k\Delta d = \frac{2\pi}{\lambda}\Delta d \qquad \Delta d = m\lambda$ Constructive

$\Delta d = \pm\frac{(2m+1)\lambda}{2}$ Destructive $m = 0, 1, 2, \ldots$



3) Properties of path (n).

$U_1 = \sqrt{I_o}\cdot e^{-jk_1 d} \qquad U_2 = \sqrt{I_o}\cdot e^{-jk_2 d}$

$I = 2I_o[1+\cos(k_2 - k_1)d]$

$(k_2 - k_1)d = \frac{2\pi}{\lambda}d(n_2 - n_1) = \frac{2\pi d}{\lambda}\Delta n$

Applications: Interferometry

1) Strain/stress
2) Index change (uniformity)
3) Layer thickness.

1) Two monochromatic plane light waves, same wavelength.

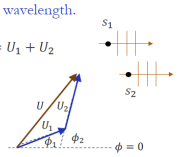
$U_1 = \sqrt{I_1}\cdot e^{j\phi_1} \qquad U_2 = \sqrt{I_2}\cdot e^{j\phi_2} \qquad U(r) = U_1 + U_2$

$I(r) = |U(r)|^2 = |U_1 + U_2|^2$

$I(r) = |U_1|^2 + |U_2|^2 + U_1U_2^* + U_1^*U_2$

$I(r) = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos(\phi_2 - \phi_1)$

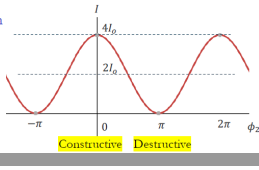
Interference equation



Example. Draw the interference pattern for two sources with $I_1 = I_2 = I_o$

$I = 2I_o[1+\cos(\phi_2 - \phi_1)]$

What is the pattern on the screen?



2) Two monochromatic plane light waves, same wavelength, at oblique angle.

$U_1 = \sqrt{I_o}\cdot e^{-jkz} \qquad U_2 = \sqrt{I_o}\cdot e^{-jk(x\sin\theta + z\cos\theta)}$

At the screen,

$U_1 = \sqrt{I_o} \qquad U_2 = \sqrt{I_o}\cdot e^{-jkx\sin\theta}$

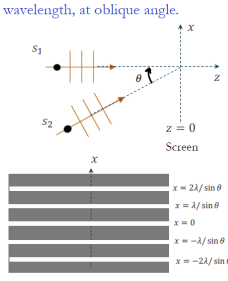
$I = 2I_o[1+\cos(kx\sin\theta)]$

$kx\sin\theta = m(2\pi)$ Maxima $m = 0, 1, 2, \ldots$

$x_{max} = \frac{m(2\pi)}{k\sin\theta} = m\frac{\lambda}{\sin\theta}$

$kx\sin\theta = (2m+1)\pi$ Minima

$x_{min} = \frac{(2m+1)\pi}{k\sin\theta} = \left(\frac{2m+1}{2}\right)\frac{\lambda}{\sin\theta}$



Applications: Lithography

Example. Calculate the required angle to produce a 1 μm feature using blue light ($\lambda = 450\ nm$).

3) Multiple wave interference. Consider M monochromatic plane light waves with same amplitude and same wavelength. Constant phase shift ϕ between each.

$U_m = \sqrt{I_o}\cdot e^{j(m-1)\phi} \qquad m = 1, 2, \ldots, M$

$U(r) = \sum_1^M U_m = \sqrt{I_o}\left(1 + e^{j\phi} + e^{j2\phi} + \cdots + e^{j(M-1)\phi}\right)$

$U(r) = \sqrt{I_o}\left(\frac{1-e^{jM\phi}}{1-e^{j\phi}}\right)$

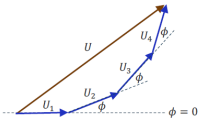
$I(r) = |U(r)|^2 = I_o\left|\frac{1-e^{jM\phi}}{1-e^{j\phi}}\right|^2 = I_o\frac{\sin^2(M\phi/2)}{\sin^2(\phi/2)}$

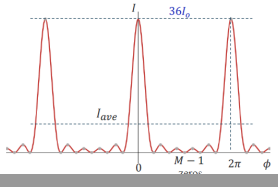
Example. Draw the interference pattern for six sources with I_o. Find the maximum and average intensity.

$I_{max} = M^2 I_o$

$I_{ave} = ?$ Assignment

Possible application: intense light





4) Two spherical monochromatic light sources, in-phase with same wavelength.

For spherical waves,

$U(r) = \frac{A}{r}e^{-jkr}$

Phase difference is related to $r_2 - r_1$

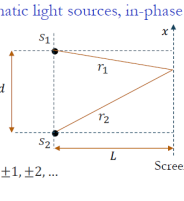
Constructive interference for $r_2 - r_1 = n\lambda \qquad n = 0, \pm1, \pm2, \ldots$

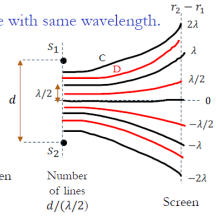
Destructive interference for $r_2 - r_1 = \frac{(2n+1)\lambda}{2}$

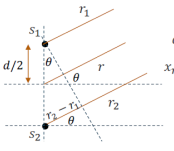
Locations of max/min on the screen (assume $r \gg d$)

Phase difference is given by

$\delta = \frac{2\pi}{\lambda}(r_2 - r_1) = \frac{2\pi}{\lambda}(d\sin\theta)$







Constructive interference

$d\sin\theta = n\lambda \rightarrow \sin\theta_n = n\lambda/d$

$x_n = L\tan\theta_n \approx L\sin\theta_n = nL\lambda/d$

Destructive interference

$x_n = \frac{(2n+1)L\lambda}{2d}$

4) Two spherical monochromatic light sources, in-phase with same wavelength.

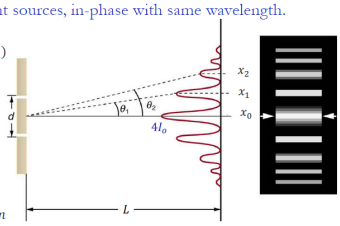
$I_n = I_o\left(\frac{\sin\delta_n}{\sin\delta_n/2}\right)^2 \qquad \delta_n = \frac{2\pi}{\lambda}(d\sin\theta_n)$

$I_n = 4I_o(\cos\delta_n/2)^2$

Example.

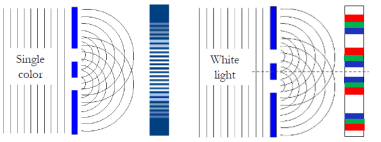
d=68 μm
λ=632.8 nm (red)
L=10 m

$\theta_o = 0, \ x_o = 0, \ \theta_1 = 0.412°, \ x_1 = 7.2\ cm$



Question. How are the two sources realized in practice?

Question. If white light is used, how does the interference pattern look like?



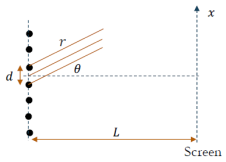
5) Many (N) spherical monochromatic light sources, in-phase with same wavelength. The Gratings problem.

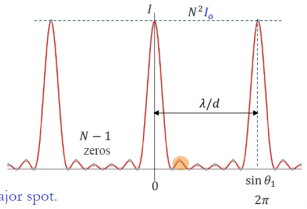
Look at the problem 2 sources at a time.

$$\delta = \frac{2\pi}{\lambda}(d\sin\theta_n) = n2\pi \qquad \text{Constructive}$$

$$\sin\theta_n = n\lambda/d \qquad x_n \approx nL\lambda/d$$

$$I = I_o\left[\frac{\sin(N\,\delta/2)}{\sin(\delta/2)}\right]^2 \qquad I_{max} = N^2 I_o \qquad \delta = 0, 2\pi, 4\pi, \ldots$$





Example.
d=2 μm
λ=532 nm (green)
Spot size=2 mm
(1000 lines)
L=5 m → Δx=1.3 m



Find the intensity of the first max after the major spot.

6) Single slit interference. The Diffraction problem.

Phase difference is related to

$$\beta = \frac{2\pi}{\lambda}(r_2 - r_1) = \frac{2\pi}{\lambda}\frac{a}{2}\sin\theta$$

Destructive interference with

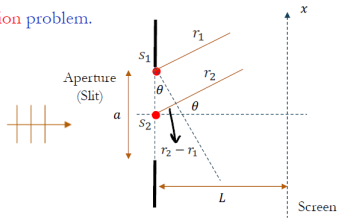
$$\frac{a}{2}\sin\theta = n\frac{\lambda}{2} \quad \rightarrow \sin\theta = n\frac{\lambda}{a}$$

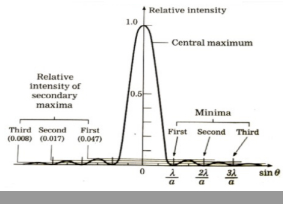
Constructive interference max at $\theta = 0$

Each two points produce the same effect.

Intensity is given by $I(\beta) \approx I_o\left(\frac{\sin\beta}{\beta}\right)^2$

$\sin\theta$	β	$\sin\beta$	I	x
0	0	0	I_o	0
λ/a	π	0	o	$\lambda L/a$
$2\lambda/a$	2π	0	o	$2\lambda L/a$





6) Single slit interference. The Diffraction problem.

Example.1. Find the spot size projected on a screen (L=3 m) by a red laser (630 nm) through an aperture of 0.1 mm.

$$x_1 = \frac{\lambda L}{a} = 2\ cm, x_{-1} = -2\ cm \qquad \text{Spot size on screen= 4 cm}$$

Example.2. A 550-nm light through an aperture produces the dark bright spot at 45°. Find the aperture size.

$$a\sin\theta_2 = 2\lambda \qquad a = \frac{2(550\ n)}{\sin 45} = 1.56\ \mu m \qquad \theta_1 = \sin^{-1}\left(\frac{\lambda}{a}\right) = 20.7°$$

Example.3. A 589-nm light goes through a 7.5-μm aperture find the first few angles of the dark spots.

$$\theta_1 = \sin^{-1}\left(\frac{\lambda}{a}\right) = 4.5° \qquad \theta_2 = 9.05° \qquad \theta_{13} = NA$$

Example.4. A 550-nm light goes through a 1-μm aperture find the first few angles of the dark spots.

$$\theta_1 = \sin^{-1}\left(\frac{\lambda}{a}\right) = 33.4° \qquad \theta_2 = NA$$

6) Single slit interference. The Diffraction problem.

The light intensity of a grating becomes

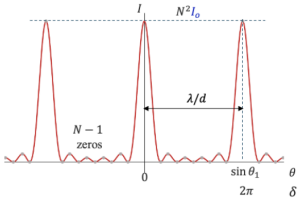
$$I = I_o\left[\frac{\sin(N\,\delta/2)}{\sin(\delta/2)}\right]^2 \cdot \left(\frac{\sin\beta}{\beta}\right)^2$$

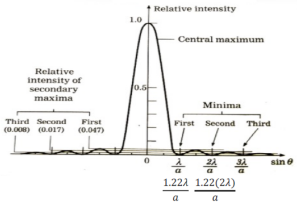
Interference Diffraction

$$\delta = \frac{2\pi}{\lambda}(d\sin\theta) \qquad \beta = \frac{\pi a}{\lambda}(\sin\theta)$$

Some bright spots in the gratings pattern can possibly be nulled by dark spots in the diffraction pattern.

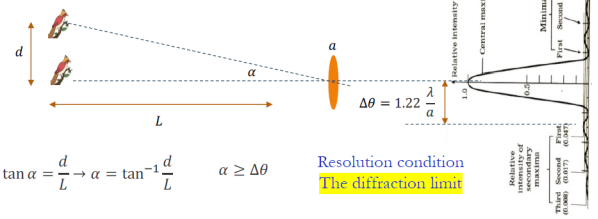
For example, if $a = d/5$, the 5th order bright spot in the grating pattern is lost because of the null in the diffraction pattern.





6) Single slit interference. The Diffraction problem.

For a circular aperture (e.g., human eye) $\frac{\lambda L}{a} \rightarrow 1.22\frac{\lambda L}{a}$



$$\tan\alpha = \frac{d}{L} \rightarrow \alpha = \tan^{-1}\frac{d}{L} \qquad \alpha \geq \Delta\theta$$

Resolution condition
The diffraction limit

For the human eye, a=4 mm, λ=500 nm $\qquad \Delta\theta = 1.5\times10^{-4}$ rad

For Hubble telescope, a=2.4 m, λ=500 nm, $\Delta\theta = 0.28\times10^{-6}$ rad

Review of the fundamental laws of electricity and magnetism.

Fields	$\vec{E} = \frac{\vec{F}_e}{q}$	$(\vec{v}\times\vec{B}) = \frac{\vec{F}_m}{q}$
Flux densities	$\vec{D} = \varepsilon\,\vec{E}$	$\vec{B} = \mu\,\vec{H}$
Gauss's Law	$\vec{\nabla}\cdot\vec{D} = \rho_v$	$\oint \vec{D}\cdot\vec{ds} = Q_{enc}$
Gauss's Law	$\nabla\cdot\vec{B} = 0$	$\oint \vec{B}\cdot\vec{ds} = 0$
Ampere's Law	$\vec{\nabla}\times\vec{H} = \vec{J}$	$\oint \vec{H}\cdot\vec{dl} = I_{enc}$
Faraday's Law	$\mathcal{E}mf = -\frac{\partial\varphi}{\partial t}$	$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$

Charge conservation
The continuity equation $\qquad \vec{\nabla}\cdot\vec{J} + \frac{\partial\rho_v}{\partial t} = 0$

$$\vec{J} = \sigma\vec{E}$$

Basic Quantities

(V/m) $\vec{E}, \vec{D}$ (C/m²)

(A/m) $\vec{H}, \vec{B}$ (Wb/m²)

(A/m²) $\vec{J}, \rho_v$ (C/m³)

Material Properties

ε, μ, σ

(F/m) (H/m) (S/m)

$$\vec{\nabla}\times\vec{H} = \vec{J} + \frac{\partial\vec{D}}{\partial t} \qquad \vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$$

$$\vec{\nabla}\cdot\vec{D} = \rho_v \qquad \nabla\cdot\vec{B} = 0$$

$$\vec{D} = \varepsilon\,\vec{E} \qquad \vec{B} = \mu\,\vec{H} \qquad \vec{J} = \sigma\vec{E}$$

Maxwell's Equations

Constitutive Relations

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} = -\mu\frac{\partial\vec{H}}{\partial t} \qquad \vec{\nabla}\times\vec{\nabla}\times\vec{E} = \vec{\nabla}(\vec{\nabla}\cdot\vec{E}) - \vec{\nabla}^2\vec{E}$$

$$\vec{\nabla}\times\vec{\nabla}\times\vec{E} = -\mu\frac{\partial(\vec{\nabla}\times\vec{H})}{\partial t}$$

$$\vec{\nabla}(\vec{\nabla}\cdot\vec{E}) - \vec{\nabla}^2\vec{E} = -\mu\frac{\partial}{\partial t}(\vec{J} + \frac{\partial\vec{D}}{\partial t})$$

$$\vec{\nabla}^2\vec{E} - \vec{\nabla}(\vec{\nabla}\cdot\vec{E}) = \mu\sigma\frac{\partial\vec{E}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2}$$

$$\vec{\nabla}^2\vec{E} = \mu\sigma\frac{\partial\vec{E}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2}$$

$$\vec{\nabla}^2\vec{H} = \mu\sigma\frac{\partial\vec{H}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

The EM Wave Equation

$$\vec{\nabla}^2\vec{E} = \mu\sigma\frac{\partial\vec{E}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\sigma\frac{\partial\vec{H}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

Consider a source-free $\rho_v = 0$ and lossless (non-conducting) $\sigma = 0$ medium

$$\vec{\nabla}^2\vec{E} = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

Write the wave equation in this form

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\vec{E} - \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} = 0 \qquad \nabla^2 U - \frac{1}{v^2}\frac{d^2U}{dt^2} = 0$$

Wave speed $\quad v = \frac{1}{\sqrt{\mu\varepsilon}}$ $\qquad$ For free space $\quad v = \frac{1}{\sqrt{\mu_o\varepsilon_o}} = 3.0\times10^8$ (m/s)

1) Electromagnetic waves exist.

2) The wave is composed of two interacting quantities E and H fields.

3) EM waves travel with the speed of light.

4) Light is an EM waves.

Consider a source-free $\rho_v = 0$ and lossless (non-conducting) $\sigma = 0$ medium

$$\vec{\nabla}^2\vec{E} = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

Time-harmonic waves are generated by time-harmonic sources.

$\vec{E}(r,t) = \vec{E}(r)\, e^{j\omega t}$ r here represents the space variable. $e^{j\omega t}$ Sinusoidal, single frequency

$$\vec{\nabla}^2\vec{E}(r) + k^2\vec{E}(r) = 0$$
$$\vec{\nabla}^2\vec{H}(r) + k^2\vec{H}(r) = 0$$

Helmholtz wave equation

$k = \omega\sqrt{\mu\varepsilon} = \frac{\omega}{v}$ Wave number (rad/m) (Real number)

Refractive index

$n = \sqrt{\mu_r\varepsilon_r}$ $v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c}{n}$ $k = \frac{n\omega}{c}$

A general expression of the electric field vector in rectangular coordinates is

$$\vec{E}(x,y,z,t) = E_x(x,y,z,t)\,\vec{a}_x + E_y(x,y,z,t)\,\vec{a}_y + E_z(x,y,z,t)\,\vec{a}_z$$

TH Wave equation $\vec{\nabla}^2\vec{E}(x,y,z) + k^2\vec{E}(x,y,z) = 0$ ➡ $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2\right)\vec{E}(x,y,z) = 0$

Consider a one-dimensional case (z) with one field component only.

$$\frac{\partial^2 E_x}{\partial z^2} + k^2 E_x = 0$$

$E_x(z) = A\, e^{-jkz} + B\, e^{jkz}$ Plane wave Solution

A and B are constants.

Put back the time-harmonic factor

$E_x(z,t) = A\, e^{j(\omega t - kz)} + B\, e^{j(\omega t + kz)}$

Wave travelling along the $+ve$ z direction

Wave travelling along the $-ve$ z direction

Take the $+z$ wave

$E_x(z,t) = E_o\, e^{j(\omega t - kz)}$

$E_x(z,t) = E_o\cos(\omega t - kz) + j\,E_o\sin(\omega t - kz)$

Solution Solution

Take the sine wave solution

$E_x(z,t) = E_o\sin(\omega t - kz)$

$\vec{E}(z,t) = E_o\sin(\omega t - kz)\,\vec{a}_x$

Same definitions for E_o, ω, k

Electric field Part of the wave

Amplitude

Phase

$\vec{E}(z,t) = E_o\sin(\omega t \pm kz)\,\vec{a}_x$ ← Field orientation/polarization

$\omega = 2\pi f$ Frequency

$k = \frac{\omega}{v}$ Wave number

Direction of wave propagation

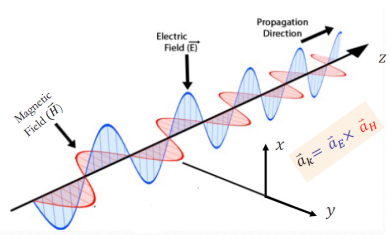
$\vec{E}(z,t) = E_o\sin(\omega t - kz)\,\vec{a}_x$

What about $\vec{H}$?

The EM fields are related using Maxwell's equations.

$\vec{E}(z) = E_o\, e^{-jkz}\,\vec{a}_x$

$\vec{\nabla}\times\vec{E} = -\mu\frac{\partial\vec{H}}{\partial t}$



For the 1D TH case, we have

$\vec{H} = \frac{1}{-j\omega\mu}\vec{\nabla}\times\vec{E} = \frac{1}{-j\omega\mu}(-jk)E_o\, e^{-jkz}\,\vec{a}_y = \frac{k}{\omega\mu}E_o\, e^{-jkz}\,\vec{a}_y$ $k^2 = \omega^2\mu\varepsilon$

$\vec{H} = \sqrt{\frac{\varepsilon}{\mu}}E_o\, e^{-jkz}\,\vec{a}_y$ $\sqrt{\frac{\mu}{\varepsilon}}$ has units of impedance (Ω) $\eta = \frac{E_o}{H_o} = \sqrt{\frac{\mu}{\varepsilon}}$ Plane wave impedance $H_o = \frac{E_o}{\eta}$

$\vec{H} = H_o\, e^{-jkz}\,\vec{a}_y$ For free-space $\eta_o = \sqrt{\frac{\mu_o}{\varepsilon_o}} = 377\ \Omega$ Other media $\eta = \frac{\eta_o}{n}$

Complex Refractive index

$n = \sqrt{\mu_r\varepsilon_r} = n' - jn''$

Complex propagation constant

$k = \frac{n\omega}{c}$

$k = \frac{\omega}{c}n'' + j\frac{\omega}{c}n'$

$k = \alpha + jk$

Field attenuation Phase constant

$\alpha = \frac{\omega}{c}n''$ $k = \frac{\omega}{c}n'$

Refractive index

$(n')^2 = \frac{\varepsilon_r'}{2} + \frac{1}{2}\sqrt{(\varepsilon_r')^2 + (\varepsilon_r'')^2}$

Extinction coefficient

$n'' = -\frac{\varepsilon_r''}{2n'}$

Other relations

$\varepsilon_r' = (n')^2 + (n'')^2$

$\varepsilon_r'' = -2n'\,n''$

Wave speed

$v = \frac{c}{n'}$

Wave impedance

$\eta = \frac{\eta_o}{n} = \frac{\eta_o}{n' - jn''}$

A general expression of the electric field vector in rectangular coordinates is

$$\vec{E}(x,y,z,t) = E_x(x,y,z,t)\,\vec{a}_x + E_y(x,y,z,t)\,\vec{a}_y + E_z(x,y,z,t)\,\vec{a}_z$$

Consider a one-dimensional case (z) with one component only.

$$\frac{\partial^2 E_x}{\partial z^2} - k^2 E_x = 0$$ k is complex

$E_x(z) = A\, e^{-kz} + B\, e^{kz}$

Wave travelling along the $+ve$ z direction

Wave travelling along the $-ve$ z direction

Complex wave impedance $\eta = \frac{E_o}{H_o} = \frac{j\omega\mu}{\gamma} = |\eta|\,\angle\theta_\eta$

$\vec{a}_k = \vec{a}_E \times \vec{a}_H$

Take the $+z$ wave

$E_x(z) = E_o\, e^{-(\alpha + jk)z}$

$E_x(z) = E_o\, e^{-\alpha z}\, e^{-jkz}$

Attenuation Phase

Take the sine wave solution

$E_x(z,t) = E_o\sin(\omega t - kz)\, e^{-\alpha z}$

$\vec{E}(z,t) = E_o\sin(\omega t - kz)\, e^{-\alpha z}\,\vec{a}_x$

$\vec{H}(z,t) = H_o\sin(\omega t - kz)\, e^{-\alpha z}\,\vec{a}_y$

Consider the following Plane wave represented by E and H.

$\vec{E} = E_o\cos(\omega t - kz)\, e^{-\alpha z}\,\vec{a}_x$

$\vec{H} = H_o\cos(\omega t - kz - \theta_\eta)\, e^{-\alpha z}\,\vec{a}_y$

$\vec{S} = \vec{E}\times\vec{H}$ Poynting vector (Wave power density W/m^2)

$\vec{S} = E_o H_o\cos(\omega t - kz)\cos(\omega t - kz - \theta_\eta)\, e^{-2\alpha z}\,\vec{a}_z$

$\vec{S} = \frac{E_o^{\,2}}{|\eta|}\cos(\omega t - kz)\cos(\omega t - kz - \theta_\eta)\, e^{-2\alpha z}\,\vec{a}_z$

The time-average power density is given by

$\vec{S}_{ave} = \frac{1}{T}\int_0^T \vec{S}\, dt\,\vec{a}_z$ (W/m^2)

For a plane wave,

$\vec{S}_{ave} = \frac{1}{2}Re[\vec{E}\times\vec{H}^*]$ ➡ $\vec{S}_{ave} = \frac{E_o^{\,2}}{2|\eta|}\cos\theta_\eta\, e^{-2\alpha z}\,\vec{a}_z$

Total average power within some area

$P_{ave} = \int_A \vec{S}\cdot\overrightarrow{dA}$ (W)

Consider an EM wave given by $\vec{E} = 4\sin(8.8\pi\times10^{14}t - kx)\,\vec{a}_z$ (V/m) travelling in a non-magnetic medium of $n = 3.8$. Find k, η and $\vec{S}_{ave}$.

Lossless medium $\alpha = 0$

Frequency $\omega = 8.8\pi\times10^{14}$ rad/s

Refractive index $n = 3.8$ (real)

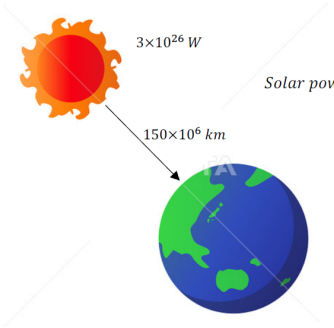
$k = \frac{\omega}{c}n = \frac{8.8\pi\times10^{14}}{3\times10^8}(3.8) = 3.5\times10^7$ rad/m

$v = \frac{c}{n} = \frac{3\times10^8}{3.8} = 0.79\times10^8$ m/s

$\eta = \frac{\eta_o}{n} = \frac{377}{3.8} = 98.6\ \Omega$

$\vec{S}_{ave} = \frac{E_o^{\,2}}{2|\eta|}\cos\theta_\eta\, e^{-2\alpha z}\,\vec{a}_x$

$\vec{S}_{ave} = \frac{4^2}{2(98.6)}\,\vec{a}_x = 81.1\,\vec{a}_x$ mW/m^2

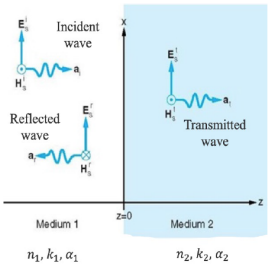


$$Total\ solar\ power = 3\times10^{26}\ W$$

$$Solar\ power\ density\ at\ Earch\ surface = \frac{3\times10^{26}}{4\pi r^2}$$

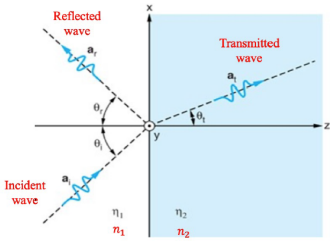
$$S_{ave} = \frac{{E_o}^2}{2|\eta|} \approx 1\ kW/m^2$$

$$E_o = 890\ V/m$$

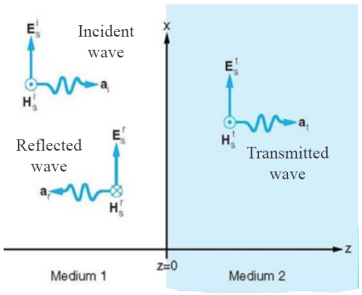


n_1, k_1, α_1 $\quad$ n_2, k_2, α_2

Normal Incidence



Oblique Incidence



n_1, k_1, α_1 $\quad$ n_2, k_2, α_2

Incident Wave

$$\vec{E}_i(z) = E_o\, e^{-jk_1 z}\, \vec{a}_x$$

$$\vec{H}_i(z) = \frac{E_o}{\eta_1} e^{-jk_1 z}\, \vec{a}_y$$

Reflected Wave

$$\vec{E}_r(z) = \Gamma\, E_o\, e^{+jk_1 z}\, \vec{a}_x$$

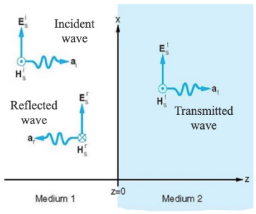
$$\vec{H}_r(z) = \frac{\Gamma E_o}{\eta_1} e^{+jk_1 z}\ (-\vec{a}_y)$$

Transmitted Wave

$$\vec{E}_t(z) = \tau E_o\, e^{-jk_2 z}\, \vec{a}_x$$

$$\vec{H}_t(z) = \frac{\tau E_o}{\eta_2} e^{-jk_2 z}\, \vec{a}_y$$

$$\eta_1 = \frac{\eta_o}{n_1} \qquad \eta_2 = \frac{\eta_o}{n_2} \qquad \eta_o = \sqrt{\mu_o/\varepsilon_o}$$



n_1, k_1, α_1 $\quad$ n_2, k_2, α_2

Apply boundary conditions at interface $z = 0$

$$E_{tangential\ (1)} = E_{tangential\ (2)}$$

$$H_{tangential\ (1)} = H_{tangential\ (2)}$$

Apply boundary conditions

$$E_{1_tan} = E_o(1+\Gamma)$$

$$E_{2_tan} = E_o\tau$$

$$H_{1_tan} = \frac{E_o}{\eta_1}(1-\Gamma)$$

$$H_{2_tan} = \frac{\tau E_o}{\eta_2}$$

Field Coefficients

Reflection coefficient $\quad$ Transmission coefficient

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2} \qquad \Gamma = \tau - 1 \qquad \tau = \frac{2n_2}{n_1 + n_2}$$

Power Coefficients

Reflectance $\quad$ Transmittance

$$R = |\Gamma|^2 \qquad R + T = 1 \qquad T = \frac{n_2}{n_1}|\tau|^2$$

Power Conservation

1) Γ and τ are in general complex. $\quad$ 2) $0 \le |\Gamma| \le 1$

$$\vec{S}^i_{ave} = \frac{n_1|E_o|^2}{2\eta_o}\,\vec{a}_k$$ Incident power density

$$\vec{S}^r_{ave} = n_1 R\frac{|E_o|^2}{2\eta_o}\,(-\vec{a}_k)$$ Reflected power density

$$\vec{S}^t_{ave} = n_1 T\frac{|E_o|^2}{2\eta_o}\,\vec{a}_k$$ Transmitted power density

Reflectance (Power)

$$R = |\Gamma|^2 = \left|\frac{n_1 - n_2}{n_1 + n_2}\right|^2$$

Transmittance (Power)

$$T = 1 - R = \left|\frac{4\, n_1 n_2}{(n_1 + n_2)^2}\right|$$

Example: Glass Reflection

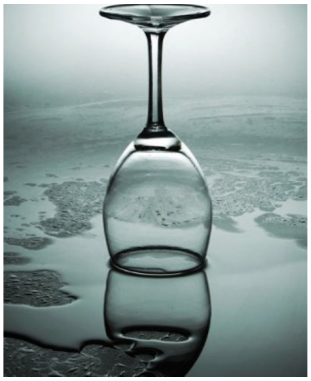
Glass refractive index, $n \approx 1.52$

$$\Gamma = \frac{1.0-1.52}{1.0+1.52} = -0.2065$$

$$R = (-0.2065)^2 = 0.0426$$

$$T = 1 - 0.0426 = 0.957$$

96% of light power goes through!

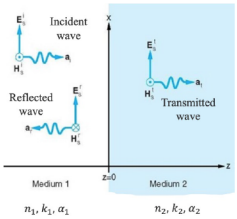


Remarks

1) Medium 1 perfect dielectric (n_1=real) and medium 2 perfect conductor ($n_2 \to \infty$)

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2} = -1$$ Total reflection

$$\tau = 0$$ Zero transmission



n_1, k_1, α_1 $\quad$ n_2, k_2, α_2

2) Medium 1 is a denser dielectric than medium 2

$n_1 > n_2 \qquad \Gamma > 0$

3) Medium 1 is a lighter dielectric than medium 2

$n_1 < n_2 \qquad \Gamma < 0$

Lossy Media

Electric Fields

$$\vec{E_i}(z) = E_o\, e^{-\alpha_1 z}\, e^{-jk_1 z}\, \vec{a}_x$$

$$\vec{E_r}(z) = \Gamma\, E_o\, e^{+\alpha_1 z}\, e^{+jk_1 z}\, \vec{a}_x$$

$$\vec{E_t}(z) = \tau E_o\, e^{-\alpha_2 z}\, e^{-jk_2 z}\, \vec{a}_x$$

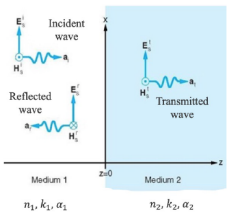
Reflection coefficient

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2}$$

Transmission coefficient

$$\tau = \frac{2n_2}{n_1 + n_2}$$

(n_1 and n_2 complex quantities)

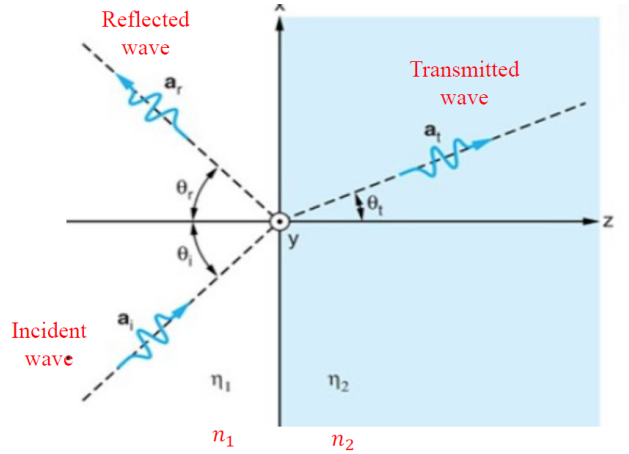


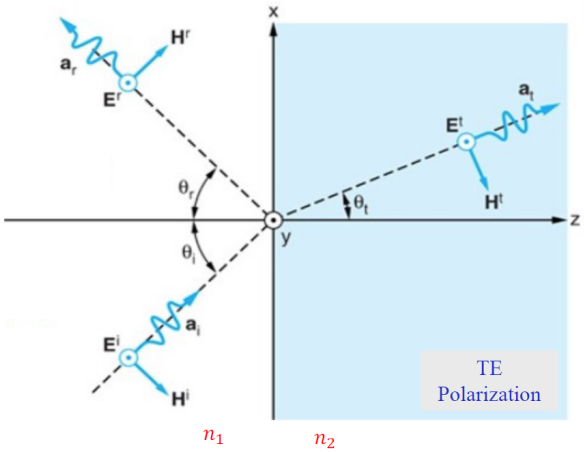
n_1, k_1, α_1 $\quad$ n_2, k_2, α_2

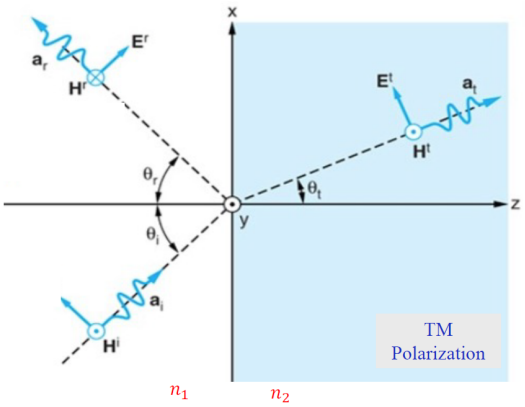
$$\vec{S}^i_{ave} = \frac{|E_o|^2}{2}\, e^{-2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right)\vec{a}_z$$

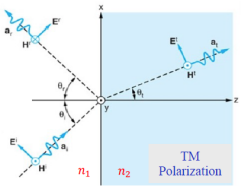
$$\vec{S}^r_{ave} = |\Gamma|^2\frac{|E_o|^2}{2}\, e^{+2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right)(-\vec{a}_z)$$

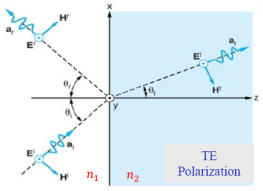
$$\vec{S}^t_{ave} = |\tau|^2\frac{|E_o|^2}{2}\, e^{-2\alpha_2 z}\, Re\left(\frac{1}{\eta_2}\right)\vec{a}_z$$











$$\Gamma_{TM} = \frac{n_1 \cos\theta_t - n_2 \cos\theta_i}{n_1 \cos\theta_t + n_2 \cos\theta_i}$$

$$\Gamma_{TM} = \frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)}$$

$$\tau_{TM} = \frac{2\, n_1 \cos\theta_i}{n_1 \cos\theta_t + n_2 \cos\theta_i}$$

$$\tau_{TM} = (1 + \Gamma_{TM}) \frac{\cos\theta_i}{\cos\theta_t}$$

Snell's Law

$$\sin\theta_t = \frac{n_1}{n_2} \sin\theta_i$$

$$R = |\Gamma|^2$$

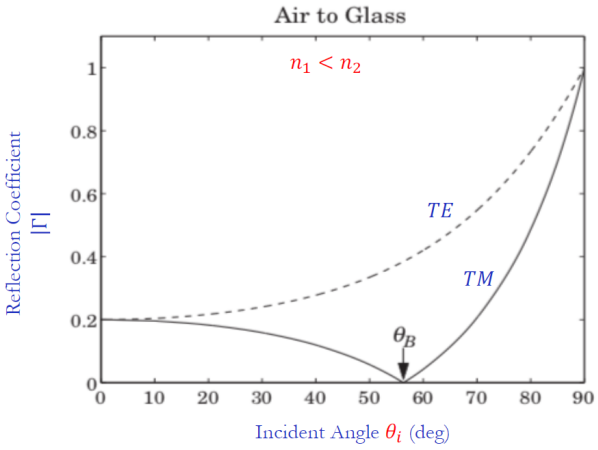
$$T = 1 - R$$

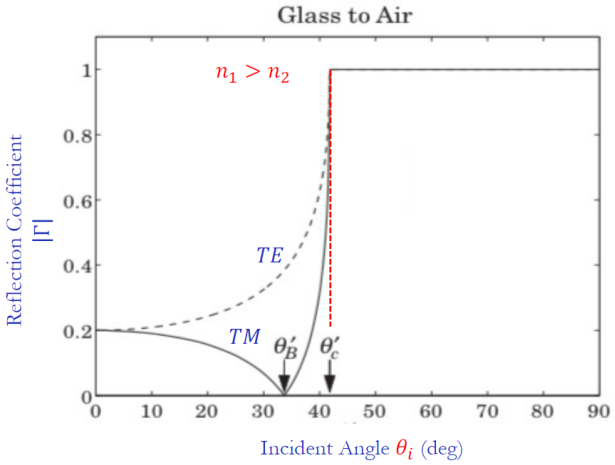
$$\Gamma_{TE} = \frac{n_1 \cos\theta_i - n_2 \cos\theta_t}{n_1 \cos\theta_i + n_2 \cos\theta_t}$$

$$\Gamma_{TE} = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)}$$

$$\tau_{TE} = \frac{2\, n_1 \cos\theta_i}{n_1 \cos\theta_i + n_2 \cos\theta_t}$$

$$\tau_{TE} = 1 + \Gamma_{TE}$$





1) Total Transmission. $|\Gamma| = 0$, Only TM polarization.

Incident Angle $\theta_i = \tan^{-1}(n_2/n_1)$ Brewster Angle θ_B

In this case $\theta_i + \theta_t = 90^o$

Air-Glass: $\theta_B = \tan^{-1}(1.52/1)$

$\theta_B = 56.7^o$

2) Total Reflection $|\Gamma| = 1$, Both TE and TM polarizations, for $n_1 > n_2$ only.

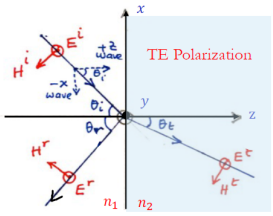
Incident Angle $\theta_i = \sin^{-1}(n_2/n_1)$ Critical Angle θ_c

Happens for any angle $\theta_i \geq \theta_c$

Glass-Air: $\theta_c = \sin^{-1}(1/1.52)$

$\theta_c = 41.1^o$

A single-frequency plane wave is incident obliquely from air onto a planar interface with a non-magnetic medium having a refractive index of $\sqrt{2}$. The electric field of the incident wave has the following phasor form $\vec{E} = [\frac{1}{\sqrt{2}}\vec{a}_y]\, e^{-jk_o(z-\sqrt{2}x)/\sqrt{3}}$ V/m



From the $\vec{E}_i$ equation, the z incident wave takes the cosine. $\cos\theta_i = \frac{1}{\sqrt{3}}$

The x incident wave takes the sine. $\sin\theta_i = \frac{\sqrt{2}}{\sqrt{3}}$

$\theta_i = 54.7°$

From Snell's law; $\sin\theta_t = \frac{n_2}{n_1}\sin\theta_i$

$\theta_t = 35.3°$

$$n_1 \sin\theta_i = n_2 \sin\theta_t$$

$$\Gamma_{TE} = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)} = -0.333$$

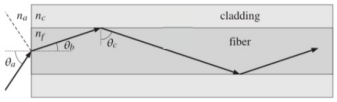
$$\tau_{TE} = 1 + \Gamma_{TE} = 0.667$$

$$\vec{H}_r = \frac{(-0.333)}{(377)(\sqrt{2})}\, e^{jk_o(\sqrt{2}x+z)/\sqrt{3}}\, (\vec{a}_x - \sqrt{2}\vec{a}_z)/\sqrt{3} \text{ A/m}$$

$$\vec{E}_t = \frac{(0.667)}{(\sqrt{2})}\, e^{-j\sqrt{2}k_o(-x+\sqrt{2}z)/\sqrt{3}}\, (\vec{a}_y) \text{ V/m}$$

The Optical Fiber.

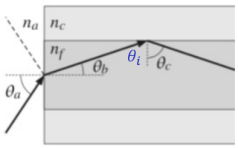
The acceptance angle θ_a should correspond to the critical angle inside the fiber at the cladding-fiber interface. $\theta_i = \theta_c = 90 - \theta_b$



Snell's Law $n_a \sin\theta_a = n_f \sin\theta_b$

$$\sin\theta_a = \frac{n_f}{n_a}\sin\theta_b = \frac{n_f}{n_a}\sin(90-\theta_c)$$

$$\sin\theta_a = \frac{n_f}{n_a}\cos\theta_c = \frac{n_f}{n_a}\sqrt{1-(\sin\theta_c)^2}$$



Critical angle condition

$$\sin\theta_c = \frac{n_c}{n_f} \Rightarrow \sin\theta_a = \frac{\sqrt{(n_f)^2-(n_c)^2}}{n_a}$$

$n_a = 1, n_f = 1.49, n_c = 1.48$

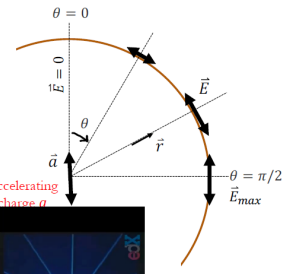
$\theta_c = 83.4°$

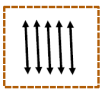
$\theta_a = 9.9°$

The polarization of the EM wave is a description of the orientation of the electric field $\vec{E}$. That, in return, is determined by the way the wave (field) is produced.

$$\vec{E} \propto q\frac{a}{r}\sin\theta\,\vec{a}_E$$

$$\vec{S} \propto q^2\frac{a^2}{r^2}\cdot\sin^2\theta\,\vec{a}_r$$

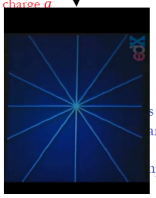




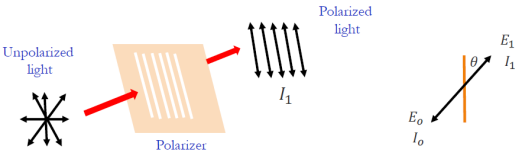
Polarized light



Unpolarized light



s of
arized light.
npolarized light?



Polarizer output
Integrate over all angles

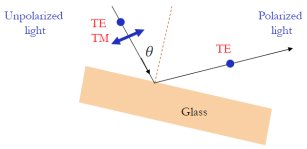
$$I_1 = \frac{1}{2}I_o$$

$$E_1 = E_o\cos\theta$$

$$I_1 = I_o\cos^2\theta$$

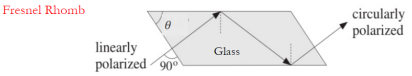
Malus' Law

1) Linear polarization.
2) Intensity goes down.
3) Light color changes??



To polarize light without a polarizer, Find the angle.

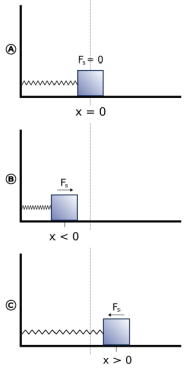
Application?



Adjust the angle of the rhomb θ such that the output beam is circularly polarized.

For glass with n=1.52, two angles can satisfy the requirement θ = 55.46° and 47.55°

Example 1. Calculate the emerging light intensity when a third polarizer is placed at 45° between two cross polarizers. The incident light is unpolarized with intensity of 600 W/m². $I_{out} = 75\ W/m^2$



$$m\frac{\partial^2\vec{x}}{\partial t^2} + m\Gamma\frac{\partial\vec{x}}{\partial t} + m\omega_0^2\vec{x} = -q\vec{E}$$

acceleration force; $m \Rightarrow m_e$ mass of an electron

frictional force; Γ damping rate (loss/sec)

restoring force; $\omega_0 = \sqrt{\frac{K}{m}}$ natural frequency

electric force

Fourier transform

$$m(-j\omega)^2\vec{x}(\omega) + m\Gamma(-j\omega)\vec{x}(\omega) + m\omega_0^2\vec{x}(\omega) = -q\vec{E}(\omega)$$

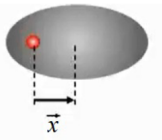
Simplify

$$\left(-m\omega^2 - j\omega m\Gamma + m\omega_0^2\right)\vec{x}(\omega) = -q\vec{E}(\omega)$$

$$\vec{x}(\omega) = -\frac{q}{m_e}\frac{\vec{E}(\omega)}{\omega_0^2 - \omega^2 + j\omega\Gamma}$$

$$\vec{P}(\omega) = N\langle\vec{p}(\omega)\rangle = \varepsilon_0\chi(\omega)\vec{E}(\omega)$$

$$\chi(\omega) = \left(\frac{Nq^2}{\varepsilon_0 m_e}\right)\frac{1}{\omega_0^2 - \omega^2 + j\omega\Gamma}$$



$$\vec{p}(\omega) = -q\vec{x}(\omega)$$

$$\chi(\omega) = \frac{\omega_p^2}{\omega_0^2 - \omega^2 + j\omega\Gamma}$$

$$\omega_p^2 = \frac{Nq^2}{\varepsilon_0 m_e}$$ plasma frequency

Lorentz Model

$q = 1.60217646\times10^{-19}$ C

$\varepsilon_0 = 8.8541878176\times10^{-12}$ F/m

$m_e = 9.10938188\times10^{-31}$ kg

$$\varepsilon_r(\omega) = 1 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 - j\omega\Gamma} \qquad \omega_p^2 = \frac{Nq^2}{\varepsilon_0 m_e}$$

$$\varepsilon_r(\omega) = \varepsilon_r'(\omega) - j\varepsilon_r''(\omega)$$

$$\varepsilon_r'(\omega) = 1 + \omega_p^2\frac{\omega_0^2 - \omega^2}{\left(\omega_0^2 - \omega^2\right)^2 + \omega^2\Gamma^2}$$

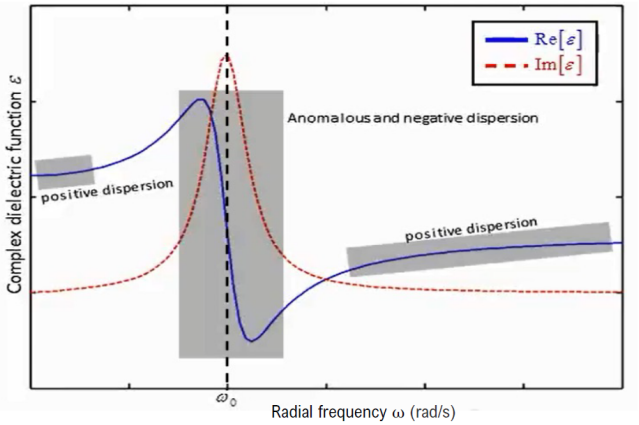
$$\varepsilon_r''(\omega) = \omega_p^2\frac{\omega\Gamma}{\left(\omega_0^2 - \omega^2\right)^2 + \omega^2\Gamma^2}$$

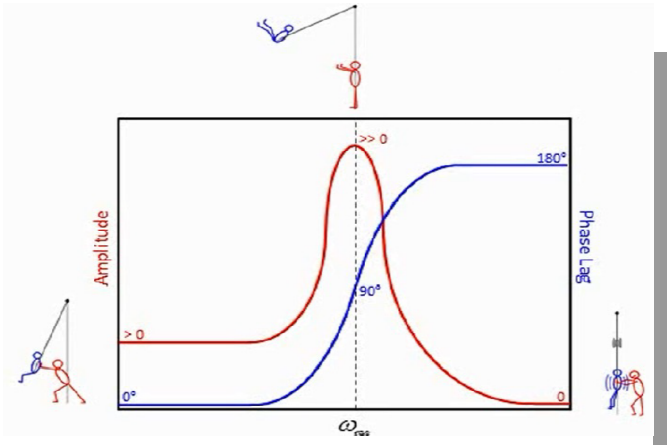
$$n = \sqrt{\mu_r\varepsilon_r} = n' - jn''$$ Complex Refractive index

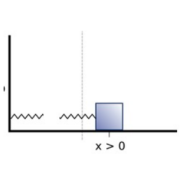
$$(n')^2 = \frac{\varepsilon_r'}{2} + \frac{1}{2}\sqrt{(\varepsilon_r')^2 + (\varepsilon_r'')^2}$$

$$n'' = -\frac{\varepsilon_r''}{2n'}$$ Extinction coefficient

$$\alpha = 2\frac{\omega}{c}n''$$ Absorption Coefficient







$$m\frac{\partial^2\vec{x}}{\partial t^2}+m\Gamma\frac{\partial\vec{x}}{\partial t}+m\omega_0^2\vec{x}=-q\vec{E}$$

acceleration force; $m \Rightarrow m_e$ mass of an electron; frictional force, Γ damping rate (loss/sec); restoring force; $\omega_0=\sqrt{\frac{K}{m}}$ natural frequency; electric force

$$\varepsilon_r(\omega)=1-\frac{\omega_p^2}{\omega^2-j\omega\Gamma}$$

$$\varepsilon_r=\left(1-\frac{\omega_p^2\tau^2}{1+\omega^2\tau^2}\right)+j\left(\frac{\omega_p^2\tau/\omega}{1+\omega^2\tau^2}\right)$$

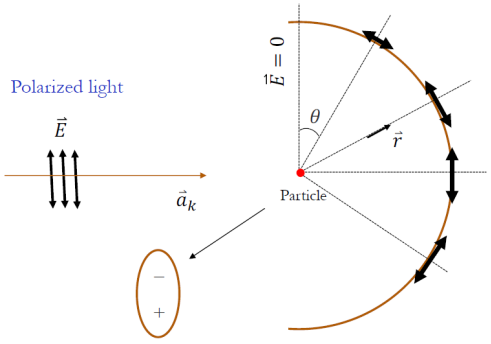
Drude Model

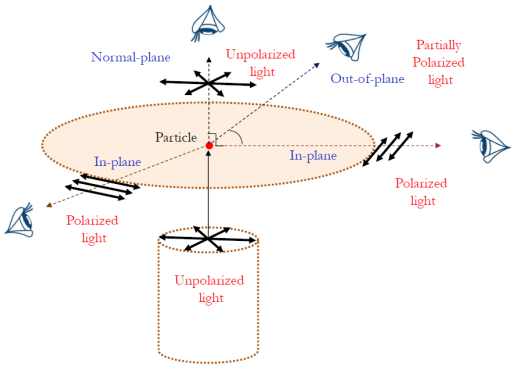
$$n=\sqrt{\mu_r\varepsilon_r}=n'-jn''$$ Complex Refractive index

$$(n')^2=\frac{\varepsilon_r'}{2}+\frac{1}{2}\sqrt{(\varepsilon_r')^2+(\varepsilon_r'')^2}$$

$$n''=-\frac{\varepsilon_r''}{2n'}$$ Extinction coefficient

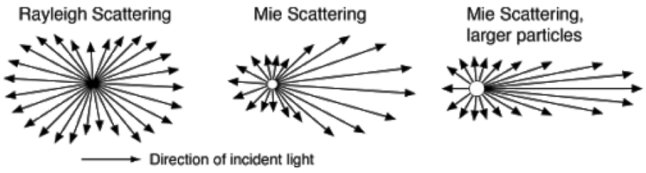
$$\alpha=2\frac{\omega}{c}n''$$ Absorption Coefficient





Scattering Forms

The wavelength-particle size relation determines the expected form of light scattering



Rayleigh Scattering

Scattering is a very strong function of wavelength.

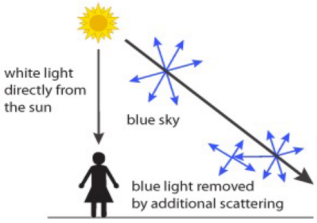
$$I\propto\frac{1}{\lambda^4}$$

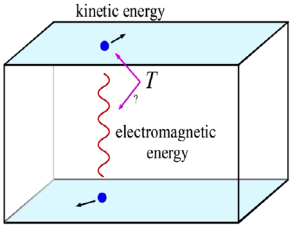
Dominant for particles of size $<\frac{1}{10}$ the incident light wavelngth.

Explains the Blue sky!

Air particle and gas molecules have size ~ 0.1 μm.

Blue light wavelength (400 nm) is about 10 times scattered than red light (wavelength around 700 nm).

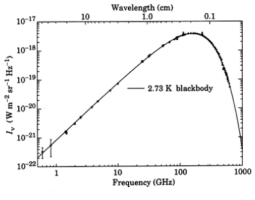




Rayleigh-Jeans law

$u(\nu)=A\nu^2kT$

(straight-line on a log-log plot)



Each mode sho Classical physics $u(\nu)=A\nu^2kT$ nergy kT

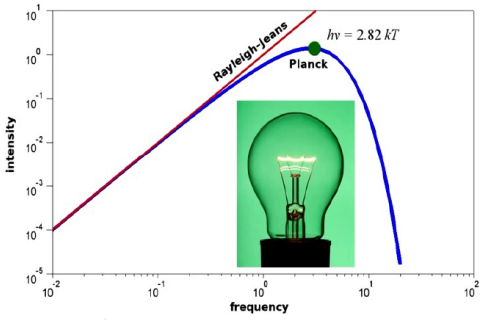
In thermodynan more modes at high frequencies $\langle E\rangle_{mode}=kT$ diation to increase with te

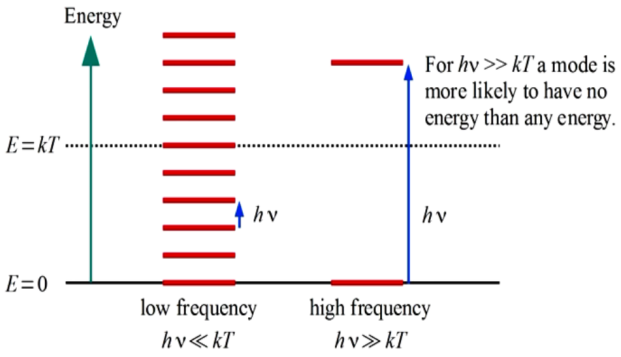
for finite h

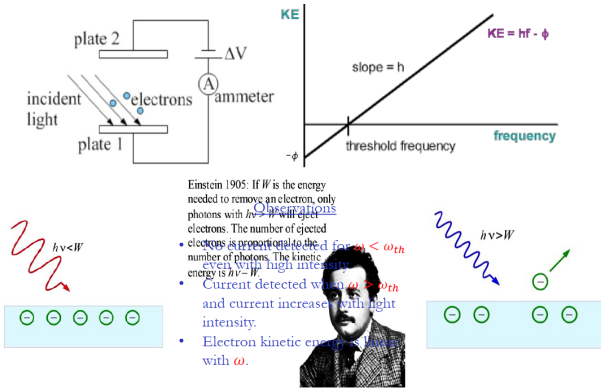
$E=nh\nu$

$$u(\nu)=A\nu^2\frac{h\nu}{e^{h\nu/kT}-1}$$ Planck's law (1900)

$h=6.626\cdot10^{-34}$Js







Einstein 1905: If W is the energy needed to remove an electron, only photons with $h\nu > W$ will eject electrons. The number of ejected electrons is proportional to the number of photons. The kinetic energy is $h\nu - W$.

Observations

- No current detected for $\omega < \omega_{th}$ even with high intensity.
- Current detected when $\omega > \omega_{th}$ and current increases with light intensity.
- Electron kinetic energy increases with ω.

Based on these observations, the photon concept emerges

What is a photon?

A light wave with frequency ω can be thought of as composed of a stream of photons (energy particles) that have the same wave speed (c) and travel in the same wave direction.

Photon Energy $E = h f = \hbar\,\omega = hc/\lambda$

Photon Momentum $p = \frac{\hbar\,\omega}{v} = \frac{h}{\lambda}$

Photon mass = 0

Photon spin = 1

Photon Stream

Average Photon Flux density in a light wave $\varphi = \frac{I}{\hbar\,\omega}$ $\frac{photons}{s \cdot m^2}$

Average Photon Flux $\int \varphi\, dA$ $\frac{photons}{s}$

Plank's constant $h = 6.626\times10^{-34}\ Js$

Reduced Plank's constant ($h/2\pi$) $\hbar = 1.055\times10^{-34}\ Js$

$\frac{photons}{s \cdot cm^2}$

Star light 10^6

Moon light 10^8

Sun light 10^{14}

Laser light 10^{22}

Example. What is the energy in a radiation mode at $\lambda = 1\ \mu m$, in free space?

$\lambda = 1\ \mu m$ ⟶ $\omega = 2\pi\, c/\lambda$

Infrared

$\omega = 2\pi\times3\times10^{14}$ rad/s

Photon Energy $E = \hbar\,\omega = 1.99\times10^{-19}\ J$

$E = \frac{1.99\times10^{-19}}{q} = 1.24\ eV$

Example. What is the photon flux in a 0.2-μm light, if the total power over some area is 1 nW?

$\lambda = 0.2\ \mu m$ ⟶ $\hbar\,\omega = 10\times10^{-19}\ J$

UV-C

Photon flux = $\frac{1\times10^{-9}}{10\times10^{-19}} = 1\times10^{9}$ $\frac{photons}{s}$

Ultraviolet light with a wavelength of 240 nm illuminates plate 1 in the figure. Consequently, electrons are emitted and driven across the gap to plate 2 and hence current is read on the ammeter. The battery voltage is gradually increased until the current drops to zero. The battery voltage at this point is 1.2 V.

Find,

1) The energy of the illuminating light.

   $E = 1.24/0.24 = 5.17\ eV$

2) Maximum kinetic energy of the electron beam.

   $KE = 1.2\ eV$

3) The work function of the metal.

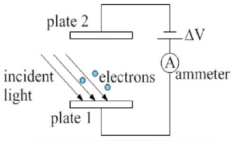
   $W = 5.17 - 1.2 = 3.93\ eV$

4) The metal used for plate 1.

   Al

5) The maximum wavelength that causes current.

   $\lambda_{max} = ch/\omega = 330\ nm$



Metal	Work Function, ϕ, (in eV)
Na	2.28
Al	4.08
Cu	4.70
Zn	4.31
Ag	4.73
Pt	6.35
Pb	4.14
Fe	4.50

Light pushes on things!

The classical (wave) concept of wave propagated power can be blended with the quantum mechanical concept to show a "classical" radiation pressure, as follows.

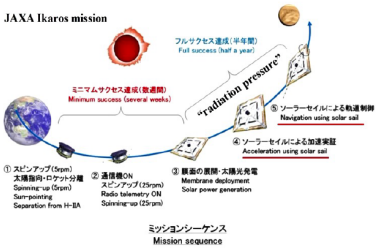
$$\frac{Power\ density\ (S_{ave})}{velocity\ (v)} = \frac{Power}{area}\frac{1}{v}$$

$$= \frac{Energy}{v}\frac{1}{area \cdot time}$$

$$= \frac{Momentum}{time}\frac{1}{area}$$

$$= \frac{Force}{area}$$

$$= Pressure$$



The solar sail